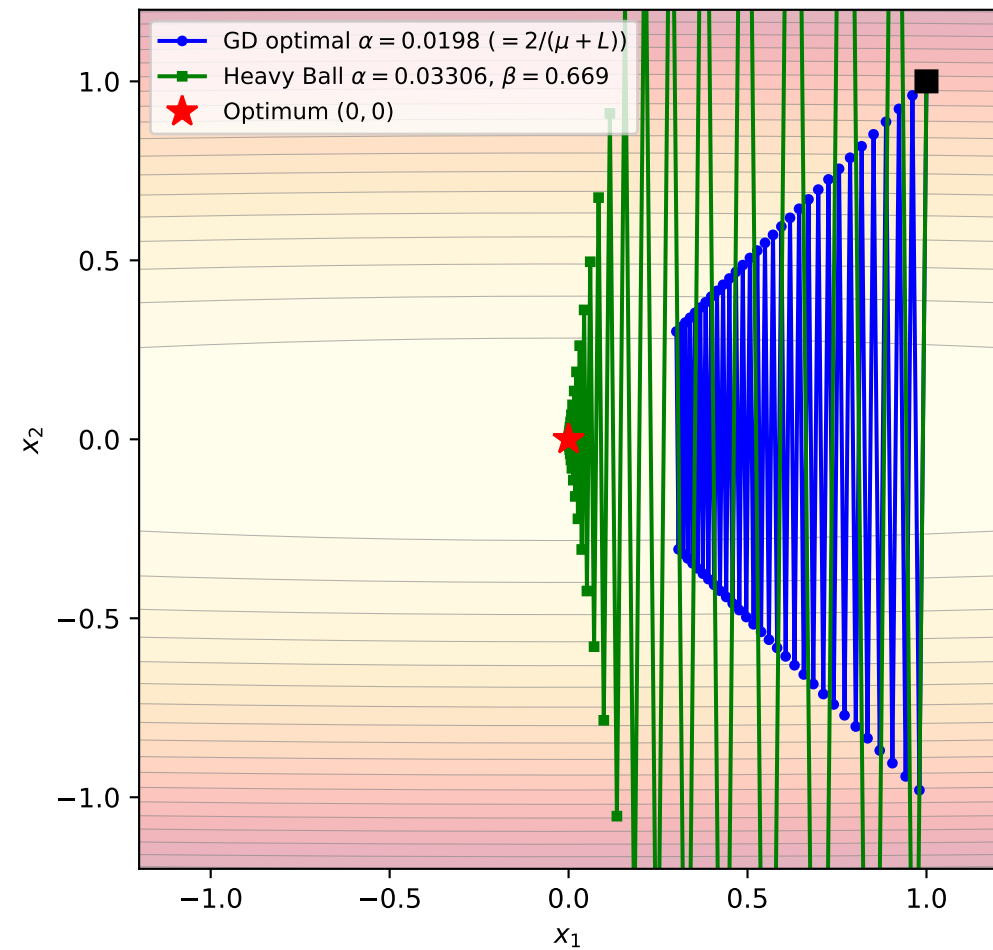


III-Conditioning ($\kappa = 100$): Zig-Zag GD vs Momentum HB

GD vs Heavy Ball on $f = (x_1^2 + 100x_2^2)/2$
($\kappa = 100$, 60 iters)



Convergence: GD vs HB ($\kappa = 100$)
HB converges $\sqrt{\kappa}$ -times faster

